

Bounded Nitsche Stabilization for Unfitted Finite Element Methods under Sliver Cuts: An Exact-Geometry Framework, a Preregistered Formula Search, and Verified Benign-Cut Validation

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Abstract

Symmetric Nitsche methods impose Dirichlet conditions on unfitted meshes through a penalty λ_T whose classical volume-scaled choice, $\lambda_T = C_k \rho(T)/h_T$ with $\rho(T) = |T \cap \partial\Omega|/|T \cap \Omega|$, diverges when the physical domain occupies a vanishing fraction of a background element (a *sliver cut*), inflating stiffness-matrix condition numbers and destroying accuracy through round-off. This work contributes three things. First, an exact-geometry computational framework in which circle cuts are clipped in closed form via Green’s theorem — carrying no tessellation error at any cut fraction — ellipses reduce affinely to the same exact path, and only genuinely non-quadratic level sets fall back to adaptive subdivision with per-cell *measured* quadrature error. Second, a preregistered protocol (formulas F1–F8, benchmark tiers T1–T5, numeric falsification gates, Bonferroni-corrected statistics) frozen before any sweep data exists, designed so that both positive and negative outcomes about bounded- λ stabilization are publishable results. Third, a fully verified implementation: 33 passing checks including closed-form circular-segment identities down to cut fraction $\sim 5 \times 10^{-7}$, agreement of moments against an independent polygon-clipping oracle, and a measured benign-cut validation in which the manufactured-solution errors converge at observed rates $p_{L^2} = 1.85$ ($R^2 = 0.995$) and $p_{H^1} = 0.97$ ($R^2 = 0.997$) for P_1 , and $p_{L^2} = 3.17$ ($R^2 = 0.9998$) and $p_{H^1} = 2.08$ ($R^2 = 0.9998$) for P_2 , with coercive spectra throughout. The preregistered controlled-sliver sweep and the hypothesis gates H1–H3 are specified in full but *not yet executed*; this manuscript reports no sliver-tier numbers because none have been measured.

1 Introduction

Let $\Omega \subset \mathbb{R}^d$ be embedded in a background domain D with a mesh $\mathcal{T}_h$ independent of $\partial\Omega$. For the Poisson problem $-\Delta u = f$ in Ω , $u = g$ on $\Gamma = \partial\Omega$, the symmetric Nitsche form seeks $u_h \in V_h$ with

$$a_h(u_h, v_h) = \sum_T \int_{T \cap \Omega} \nabla u_h \nabla v_h - \sum_T \int_{\Gamma_T} [(\partial_n u_h) v_h + u_h (\partial_n v_h)] + \sum_T \lambda_T \int_{\Gamma_T} u_h v_h = \sum_T \int_{T \cap \Omega} f v_h + \int_{\Gamma_T} g (\lambda_T v_h - \partial_n v_h) \quad (1)$$

where $\Gamma_T = T \cap \Gamma$. Discrete coercivity requires, through local trace-inverse estimates, a penalty scaling like $\lambda_T \geq C \rho(T) h_T^{-1}$ with $\rho(T) = |\Gamma_T|/|T \cap \Omega|$ (Cui and Kanschat, 2025). Under a

*Reproducibility repository: <https://github.com/mohamedhossammohamed/nitsche-cutcell-methodology>. Pre-registration SHA-256 b5f108ebad0465e7599a39726b301210946bc6198fc7054dfba615652108721f, frozen 2026–08–25 before collection of any sweep data.

sliver cut, $|T \cap \Omega| \rightarrow 0$ while $|\Gamma_T|$ stays bounded below, so $\rho(T)$ and hence λ_T diverge: the condition number inflates, and round-off swamps discretization error long before asymptotic regimes are reached. Established remedies modify the *space*: cell aggregation constructs extended finite element spaces whose condition number provably matches body-fitted behaviour (Badia et al., 2019), and related agglomeration ideas appear across method families (Rangarajan and Sukumar, 2025); alternatively, stabilization terms such as ghost penalties or polynomial extensions restore cut-position independence without changing the space (Burman et al., 2023; Zheng et al., 2026), and solver-side treatments attack the conditioning downstream (Cui and Kanschat, 2025; Voet et al., 2026).

The open question this study addresses, fixed in advance and verbatim, is: does there exist a *closed-form* stabilization parameter $\Phi(\rho, h_T, k, \kappa, a_T)$ that (a) remains bounded in the sliver limit, (b) preserves discrete coercivity independent of the cut fraction ε and mesh size, (c) preserves the optimal rate $O(h^k)$, and (d) restores the standard conditioning rate $\kappa(A_h) = O(h^{-2})$ — and does it compete with aggregation and ghost penalties on identical benchmarks?

Two properties distinguish this work from prior numerical studies of that question. First, the geometric quantities entering any candidate formula — $\rho(T)$, curvature κ , aspect ratio a_T — are computed *exactly* for the circle family rather than approximated, eliminating quadrature contamination of the conditioning signal at the very cut fractions under study (§2). Second, the entire experimental protocol was frozen and hashed before a single sweep datum existed (§3): candidate list, grids, seeds, falsification gates, and statistical corrections cannot drift after seeing data. At the time of writing the implementation has passed its full verification suite and the preregistered benign-cut validation has been measured and is reported here (§4); the controlled-sliver tier and hypothesis gates are specified but not yet run, and we state that plainly rather than implying more.

Our contributions are: (i) an exact-geometry cut-cell engine with machine-precision areas, boundary measures, and curved-boundary quadrature for circular arcs at arbitrary sliver fractions; (ii) a degree-exact collapsed-coordinate simplex rule and a P1/P2 unfitted symmetric Nitsche assembler with active-space restriction; (iii) a hashed preregistration of the complete formula search with numeric falsification criteria; (iv) measured benign-cut validation achieving optimal rates for both degrees; (v) an account of defect classes caught by independent-oracle verification, several of which produce silently wrong numerics that mass-based sanity checks do not detect.

2 Method

2.1 Exact clipping of circle cuts

For a disk boundary $\phi(x) = r^2 - |x - c|^2 > 0$ inside, the region $T \cap \Omega$ within a triangle decomposes into polygonal runs along ∂T plus circular-segment crescents. Areas follow in closed form from Green’s theorem: each directed line piece contributes $\frac{1}{2}(a_x b_y - b_x a_y)$, and an arc swept from angle a_0 to a_1 about centre c contributes

$$\frac{R}{2} \left[c_x (\sin a_1 - \sin a_0) - c_y (\cos a_1 - \cos a_0) + R(a_1 - a_0) \right], \quad (2)$$

with signed sweep taken in the traversal direction. Boundary measures $|\Gamma_T|$ are arc lengths, likewise exact. Interior integrals use a Gauss rule on the polygonal fan plus tensor Gauss strips over each crescent in a (θ, ρ) frame bounded below by the chord; angular panels exist purely for quadrature order, so combined mass reproduces the closed-form area to relative 10^{-13} while every basis-product integral converges exponentially. Because a circle may cross a convex element up to six times, the

loop construction pairs every exit crossing with the next entering crossing counter-clockwise and asserts all boundary arcs subtend less than π .

Ellipse cuts ride the same exact path: the affine map $y = D^{-1}(x - c)$ sends the ellipse to the unit disk and the element to another triangle; weights pick up $|\det D|$, normals pull back through D^{-T} , and the physical boundary measure integrates directly against elliptic speed factors. Only genuinely non-quadric shapes (e.g. superellipse ensembles) use adaptive midpoint subdivision with tangent-line clipping of mixed leaves, terminated by a curvature criterion; their accuracy is never assumed but re-measured by an internal refinement self-check logged alongside results.

2.2 Quadrature

Simplex rules are generated from tensor Gauss–Legendre products via collapsed coordinates, $(s, t) \mapsto (s, (1 - s)t)$ on the reference triangle. The Jacobian factor $(1 - s)$ raises the outer integrand degree by one, so an order- n rule is exact through total degree $2n - 2$ — one order below the naive product bound, a distinction our exactness tests verify monomial-by-monomial against closed factorial forms.

2.3 Assembly and the active space

The P1/P2 assembler implements form (1) with per-element $\lambda_T = C_k \Psi / h_T$, $C_k = 4(k + 1)^2$. Basis functions carry analytic gradients (verified against finite differences); degrees of freedom are vertices, then edge midpoints offset by the vertex count. Elements wholly outside Ω contribute nothing, and dofs supported wholly outside are dropped: the linear system lives on the active space V_h^Ω , which is essential — unrestricted systems possess exactly singular rows and columns. Optional ghost-penalty terms $\sigma_{GP} \sum_F h_F^{2k+1-d} \int_F [\partial_n u][\partial_n v]$ and AGFE-style aggregation with static condensation are implemented behind flags for the later hypothesis tiers.

3 Preregistered protocol

Before any sweep data existed, the following were frozen and committed (SHA-256 in the title footnote): thirteen closed-form formula variants F1–F6b (baseline ρ ; hard clips; aspect-ratio clip; harmonic blends; logarithmic damping; curvature-augmented), the aggregation hybrid F7 ($\varepsilon_c = 10^{-3}$), and the train-split-fitted family F8; benchmark tiers T1 (controlled sliver, ε from 0.5 to 10^{-6} crossed with five mesh sizes, two degrees, cap-height placement solved to 10^{-15}), T2 (200 seeded random realisations per mesh, frozen 70/30 split), T3 (curvature sweep), T4 (3D generalization with frozen coefficients), T5 (sign-changing coefficient); a three-solver measurement trio (sparse LU, CG+Jacobi, CG+AMG); shift-invert Lanczos spectra with explicit non-convergence flags; and numeric falsification gates H1–H3 with Bonferroni-corrected statistics. Four deviations from the originating methodology document are logged inside the preregistration itself (PETSc unavailability, supplementary mass-normalized eigenvalue logging, deferred dual-backend cross-check, and the subdivision-quadrature policy). Any protocol change after data would constitute a new registration, never a silent edit.

4 Experiments

4.1 Verification suite

Thirty-three automated checks pass. The sliver family is validated against closed-form circular segments: placing the chord at depth d makes $|T \cap \Omega| = \frac{1}{2}(\theta - \sin \theta)$ and $|\Gamma_T| = \theta$, $\theta = 2 \arccos d$, and the computed quantities match to 10^{-12} relative down to $d = 0.999999$ (cut fraction $\sim 5 \times 10^{-7}$). Random cuts are checked against an independent midpoint-shifted polygon-clipping oracle for area, first, and second moments; the subdivision path is cross-validated against the exact path where they overlap mathematically (superellipse exponent 2 is an ellipse).

This verification campaign earned its cost by catching defects that mass-consistency checks alone do not reveal: quadrature nodes sampling a square under triangle weights (masses right, moments wrong); a spurious factor one-half from double-counting the reference-triangle measure; P2 edge-dof ids colliding with vertex ids through a missing namespace offset; an absolute boundary-classification tolerance that degrades as refinement shrinks elements; and, notably, two defects inside the verification oracle itself (a sign-flipped ray parameter and a transposed $\int xy$ shoelace coefficient), caught by adjudicating against closed forms. We document these classes because each produces plausible-looking output.

4.2 Measured benign-cut validation

The single completed run reported here executes the preregistered validation tier: disk of radius 0.35 centred in the unit square (well-cut configurations), formula F1, manufactured solution $u = \sin(\pi x) \cos(\pi y)$, direct sparse-LU solve, four meshes per degree. Table 1 lists the measured errors and condition numbers exactly as logged; slopes come from OLS on log-log errors with the coefficient of determination shown.

Table 1: Measured validation results, run **98bebc59** (benign cuts, $\Omega = B_{0.35}(0.5, 0.5)^2$ inside the unit square, F1, sparse LU). Every value below is transcribed from the persisted run log.

k	n	$\ u - u_h\ _{L^2(\Omega)}$	$ u - u_h _{H^1(\Omega)}$	$\kappa(A_h)$	dropped dofs
1	8	1.200005×10^{-2}	2.319531×10^{-1}	4.21×10^2	40
1	16	4.354703×10^{-3}	1.329871×10^{-1}	4.57×10^3	152
1	32	9.554551×10^{-4}	6.404962×10^{-2}	1.30×10^5	608
1	64	2.788674×10^{-4}	3.157865×10^{-2}	8.52×10^5	2488
2	8	5.420032×10^{-4}	2.799063×10^{-2}	1.73×10^4	148
2	16	5.876154×10^{-5}	6.312563×10^{-3}	3.99×10^6	584
2	32	6.139252×10^{-6}	1.505439×10^{-3}	6.98×10^{11}	2384
2	64	7.537187×10^{-7}	3.682530×10^{-4}	6.47×10^{10}	9852

Observed rates: for $k = 1$, $p_{L^2} = 1.8470$ ($R^2 = 0.994787$) and $p_{H^1} = 0.9684$ ($R^2 = 0.996617$); for $k = 2$, $p_{L^2} = 3.1729$ ($R^2 = 0.999759$) and $p_{H^1} = 2.0812$ ($R^2 = 0.999837$). Both degrees are coercive at every mesh (shift-invert Lanczos returned positive smallest eigenvalues with explicit convergence flags). Two honest observations. First, even these “well-cut” configurations contain moderately small cuts — the circle grazes some cells — which is precisely why raw $\kappa(A_h)$ for $k = 2$ is large and non-monotone between $n = 32$ and $n = 64$: raw smallest eigenvalues are basis-dependent, which is why the preregistration logs mass-normalized companions alongside the gate quantity. Second, inactive-dof counts grow with refinement as expected from the geometry of the exterior region.

4.3 What is not yet measured

No controlled-sliver, random-ensemble, curvature, sign-changing, or three-dimensional results exist at the time of writing; tiers T1–T5 and the falsification gates H1–H3 therefore appear in this manuscript only as registered protocol. We decline to extrapolate how any bounded formula will behave at $\varepsilon \leq 10^{-3}$: the point of the frozen gates is that the data will speak first.

5 Related work

Aggregation-based cures modify the discrete space itself: Badia et al. (2019) construct aggregated unfitted spaces whose condition number provably matches body-fitted behaviour regardless of cut position, establishing the benchmark our hybrid arm must beat. Element-agglomeration strategies extend beyond Lagrangian spaces, e.g. shape-agnostic agglomeration for virtual elements (Rangarajan and Sukumar, 2025). Space-preserving stabilizations instead repair the bilinear form: polynomial-extension reconstructions remove small-cut ill-conditioning in unfitted hybrid high-order methods (Burman et al., 2023), and kernel stabilization plus ghost penalties deliver cut-position-independent constants in mixed extended virtual elements (Zheng et al., 2026). Solver-oriented work treats the symptom downstream: multiplicative vertex-patch smoothers with ghost-penalty-aware transfer give a practical multigrid for Nitsche-discretized CutFEM (Cui and Kanschat, 2025), and deflation preconditioning addresses immersed-method conditioning directly (Voet et al., 2026). Geometry-conforming immersed spaces offer an orthogonal route that replaces the cut-cell polynomial space entirely (Lin et al., 2026). The present study is complementary to all of these: it holds the space fixed, varies only the closed-form penalty Φ under preregistered falsification gates, and supplies an exact-geometry measurement bed in which the regressors entering any Φ carry no quadrature error for the quadric family.

6 Conclusion

We built and verified the measurement instrument before using it, froze the experiment before running it, and report exactly what has been measured: optimal-rate, coercive benign-cut validation for P1 and P2 unfitted symmetric Nitsche on an exact-geometry stack. The preregistered sliver sweep — thirteen formula variants, five cut fractions, five meshes, two degrees, three solvers, and the H1–H3 gates — remains to be executed; its outcome, positive or negative, will be written against the same hash that stamps this protocol. If no bounded closed-form formula survives its gates, that negative result, with evidence and exact statistics, is the publishable answer.

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